

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

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ABSTRACT

Music continuation is not reducible to similarity: the loop that best matches a session is not necessarily the one that helps a musician move it forward. Pocket Producer operationalizes this distinction inside the Audiotool DAW: it fingerprints the live session via the Nexus SDK (tempo, CLAP embeddings of session audio) and ranks Freesound Loop Dataset candidates with a five-signal fusion trained on pack co-membership weak labels—proxy supervision available before any interaction data exists—selectable alongside a transparent rules default. Offline evaluation shows the apparent ranking gain changes substantially with weak-label construction, negative regime, and embedding representation. The system therefore delivers each recommendation as an inspectable, reversible *proposal*, backed by purpose-built interaction machinery: in-context preview with per-signal evidence chips, insertion as a single undoable transaction, and an undo that also retracts the candidate’s influence on subsequent ranking.

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the question is not “what sounds similar?” but “what would be useful as the next step?” Loop compatibility [1–4], stem coherence [5–7], session-conditioned generation [8], and reversible co-creation [9] all address parts of this problem. We define **continuation utility** as the property that a candidate helps the musician make progress in the current session; unlike similarity it is asymmetric and context-dependent.

We contribute: (1) a problem formulation that separates continuation utility from similarity and compatibility; (2) a tiered negative-regime evaluation protocol with a pre-committed held-out split, yielding evidence that ranking conclusions drawn from a weak proxy shift with label construction, negative regime, and embedding representation; (3) a production DAW integration engineered for revisability: live session fingerprinting over the Nexus SDK, per-signal evidence, single-transaction insertion whose undo also retracts the candidate from the ranking context, and feedback logging.

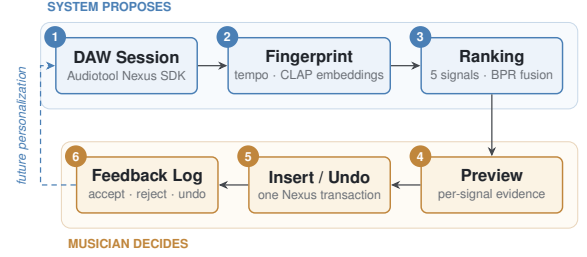


Figure 1. Pocket Producer pipeline: the system proposes (steps 1–3); the musician decides (steps 4–6).

2. SYSTEM

2.1 Session Fingerprinting

The live Audiotool session (Figure 1) is read via the Nexus SDK, which exposes the project’s devices, regions, and referenced audio samples in real time. The system extracts the project tempo from the document metadata and collects up to eight unique audio samples referenced by regions in the arrangement. Each sample is embedded with CLAP [10]; the resulting vectors form the session’s timbral fingerprint. When no session audio is available (empty project or no sample-backed regions), the system falls back to non-audio signals only.

2.2 Multi-Signal Ranking

Five signals score each FSLD [11] candidate against the session fingerprint: (i) mean-pooled and (ii) max-pooled CLAP cosine similarity to session samples, (iii) tempo proximity, (iv) key compatibility, and (v) tag Jaccard overlap between candidate tags and tags from the session’s audio fragments. A logistic fusion trained with BPR loss [12] on Freesound pack co-membership weak labels (Section 3) combines these into a single score. A hand-tuned *rules* baseline over non-audio session signals serves as the default ranker; the fusion is selectable and falls back to rules when no session audio is available.

2.3 Proposal Interface

Pack co-membership is a proxy for timbral compatibility, not continuation utility; arrangement context, creative intent, and genre conventions live outside any offline label. The interface is engineered so that this gap costs the musician nothing: every recommendation arrives as a proposal



that can be auditioned, interrogated, and revoked, with the musician holding final creative authority at every step. Each candidate is surfaced for in-context preview with per-signal evidence: reason chips naming the signals that favored it (timbral closeness to session regions, tempo fit, shared tags), so the musician can assess *why* the system ranks it highly and judge whether those reasons apply to their current goal; signals that contribute nothing emit no evidence. An accepted candidate is inserted via a single Nexus transaction; *undo* reverts the insertion and retracts the candidate’s tags from the ranking context, so subsequent recommendations are not biased by the undone choice. Acceptance, rejection, and undo events are logged for future personalization (not yet evaluated).

2.4 Deployed Model

The deployed ranker follows the same five-signal recipe but is a separate production model with four stated deltas: MS-CLAP embedding space (offline evaluation uses LAION-CLAP Music); source-weighted BPR ($\text{hard_similar} \times 2.0$, mean over 5 seeds); key constant zero online (Audiotool documents lack key-signature metadata); tag context drawn from session fragments rather than pack siblings.

3. WEAK-LABEL FUSION

From the FSLD [11] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding items grouped only by uploader. Each eligible pack with at least three items yields one leave-one-out query per item, with same-pack siblings as positives and cross-pack items as negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest cosine to the positive from another pack). A 121-source held-out split (seed 20260810) was committed to version control together with the evaluation protocol *before* any evaluation ran. Development uses only the 484 train sources; the 171 with ≥ 3 items form queries (981 examples), the rest contribute negatives only. LAION-CLAP Music [13] (512-dim) provides the embedding backbone. On the train side (mean over 5 source-grouped splits), frozen cosine reaches 0.929 overall and 0.814 on *hard_similar*; five-signal fusion reaches 0.925 and 0.813. Under pack-only labels the fusion advantage is gone: -0.03 pp on *hard_similar*, and fusion trails cosine on *hard_tempo_key* by 2.1 pp on average (negative in all 5 splits). These development-set numbers are descriptive; inference is reserved for the held-out analysis (Section 4).

4. OBSERVATIONS

The weak label measures pack co-membership, not continuation utility; the observations below chart where the proxy breaks down—and the feedback log of Section 2.3 is designed to eventually ground ranking in in-session evidence instead. Four lines of evidence jointly motivate treating the fusion as a proposal rather than a decision. **Label sensitivity.** An earlier corpus treated same-uploader items as

co-members (26% of examples, 949 sources); fusion appeared $+4.0$ pp over frozen cosine on *hard_similar*. Restricting to true pack co-membership erases the gain (Section 3).

Regime dependence. Leave-one-signal-out ablation on the dev set shows tempo helps on *hard_similar* (-0.96 pp when removed) but hurts on *hard_tempo_key* ($+0.68$ pp); signals switch sign across negative regimes, so a fused score averaging over them cannot be uniformly trusted. **Held-out uncertainty.** On a pre-committed 121-source split, the fusion–cosine advantage on *hard_similar* is $+1.5$ pp (bootstrap 95% CI $[-1.5, +4.7]$, crossing zero) under LAION-CLAP; under MS-CLAP with source-weighted BPR (a sensitivity re-analysis on the same split) it is $+9.2$ pp (CI $[+4.3, +14.3]$)—a 7.7 pp cross-configuration swing. A dev-set 2×2 factorial attributes most of the fusion–cosine delta shift to the embedding space ($+1.9$ pp) rather than the training recipe ($+0.5$ pp). **Pilot divergence.** Nine raters on 20 blind A/B pairs (Fleiss’ $\kappa = 0.294$; one pair excluded for a “neither” majority) show a descriptive divergence between weak-label signal rankings and human consensus.

5. CONCLUSION

Pocket Producer operationalizes session-conditioned retrieval as a user-controlled proposal in a production DAW. It reads the live session via the Nexus SDK, ranks candidates with a multi-signal fusion, and surfaces per-signal evidence for preview; every insertion is reversible and every interaction is logged. Since offline evaluation shows the fusion advantage is sensitive to label construction, embedding space, and negative regime, the default ranker is a transparent rules baseline; the fusion serves as a selectable alternative. Evaluating whether preview and undo improve creative outcomes, and personalization from logged feedback, is future work.

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